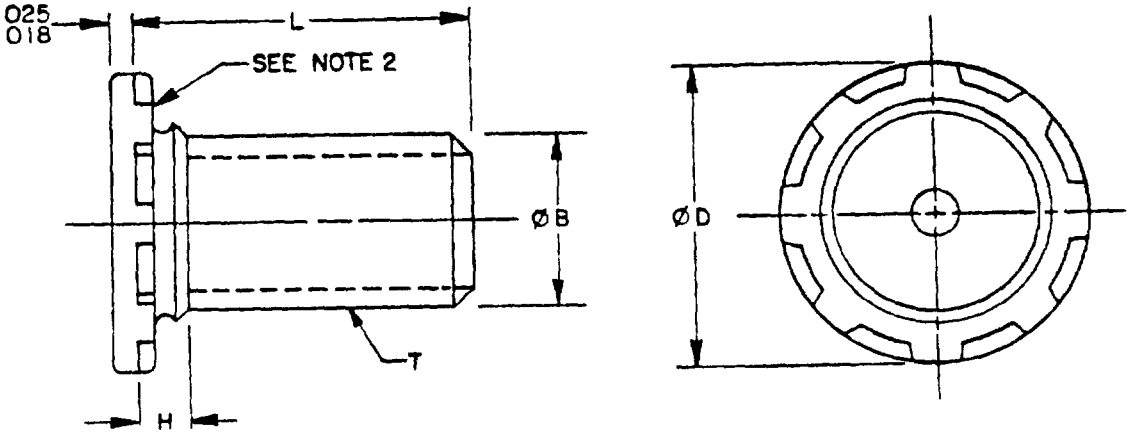
MILITARY SPECIFICATION SHEET

# STUD, SELF-LOCKING, NON-FLUSH, RIBBED CLINCH TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the studs described herein shall consist of this specification and the latest issue of MIL-S-63540.




NOTES:

1. All dimensions are in inches.
2. Clinching profile may vary.

FIGURE 1. STUD, SELF-LOCKING, NON-FLUSH, RIBBED CLINCH TYPE.



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TABLE I. PART DIMENSIONS

| T<br>THREAD SIZE<br>CLASS 2A | ØB<br>STUD<br>DIA | ØD<br>±.015 | H<br>MAX |
|---|---|---|---|
| .086-56 UNC<br>(2-56) | .086 | .141 | .070 |
| .112-40 UNC<br>(4-40) | .112 | .176 | .070 |
| .138-32 UNC<br>(6-32) | .138 | .203 | .070 |
| .164-32 UNC<br>(8-32) | .164 | .234 | .070 |
| .190-24 UNC<br>(10-24) | .190 | .250 | .090 |
| .190-32 UNF<br>(10-32) | .190 | .250 | .090 |



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TABLE II. THREAD SIZE AND LENGTH DASH NUMBERS

| L ±.015 | MATERIAL | T THREAD SIZE CLASS 2A | | | | | |
|---|---|---|---|---|---|---|---|
| | | .086-56 UNC | .112-40 UNC | .138-32 UNC | .164-32 UNC | .190-24 UNC | .190-32 UNF |
| | | THREAD SIZE AND DASH NUMBERS | | | | | |
| .250 | Carbon Steel | 1 | 6 | 12 | 21 | — | — |
| | CRES | 1C | 6C | 12C | 21C | — | — |
| .313 | Carbon Steel | 2 | 7 | 13 | 22 | — | — |
| | CRES | 2C | 7C | 13C | 22C | — | — |
| .375 | Carbon Steel | 3 | 8 | 14 | 23 | 31 | 39 |
| | CRES | 3C | 8C | 14C | 23C | 31C | 39C |
| .500 | Carbon Steel | 4 | 9 | 15 | 24 | 32 | 40 |
| | CRES | 4C | 9C | 15C | 24C | 32C | 40C |
| .625 | Carbon Steel | 5 | 10 | 16 | 25 | 33 | 41 |
| | CRES | 5C | 10C | 16C | 25C | 33C | 41C |
| .750 | Carbon Steel | — | 11 | 17 | 26 | 34 | 42 |
| | CRES | — | 11C | 17C | 26C | 34C | 42C |
| .875 | Carbon Steel | — | — | 18 | 27 | 35 | 43 |
| | CRES | — | — | 18C | 27C | 35C | 43C |
| 1.000 | Carbon Steel | — | — | 19 | 28 | 36 | 44 |
| | CRES | — | — | 19C | 28C | 36C | 44C |
| 1.250 | Carbon Steel | — | — | 20 | 29 | 37 | 45 |
| | CRES | — | — | 20C | 29C | 37C | 45C |
| 1.500 | Carbon Steel | — | — | — | 30 | 38 | 46 |
| | CRES | — | — | — | 30C | 38C | 46C |





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Requirements:

1. Material:
   a. Carbon steel shall be Grade 1005 (UNS G10050) thru Grade 1018 (UNS G10180) in accordance with ASTM A29 or ASTM A510.

   b. Corrosion resistant steel shall be Type XM-7 (UNS S30430) and 302 (UNS S30200) in accordance with ASTM A493 or Types 303 (UNS S30300) and 303Se (UNS S30323) in accordance with ASTM A582.

2. Protective Finish: Carbon steel studs shall be plated with electrodeposited cadmium in accordance with QQ-P-416, Type II, Class 3.

3. Surface Treatment: Corrosion resistant steel studs shall be cleaned and descaled in accordance with ANSI/ASTM A380 and passivated in accordance with QQ-P-35.

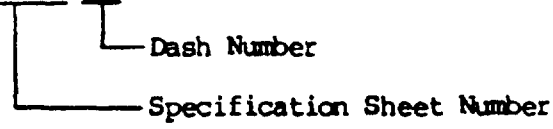
4. Hardness:
   a. Carbon steel studs, after carburizing and tempering or carbonitriding, shall have a hardness of 75-87HR15N. The minimum effective depth of case shall be .010 inch.

   b. Corrosion resistant steel studs shall have a hardness of 171HB minimum.

5. Push-out and Torque-out values: When the stud is properly installed (see installation procedures), push-out and torque-out values shall be as specified in Table III.

6. Part Number: The part number consists of the letter "M", the basic number of this specification sheet and a dash number taken from Table II.

Example: M63540/6-7C




M63540/6-7C Indicates- Stud, Self-locking, Non-flush, Ribbed Clinch Type, CRES, .112-40 UNC-2A Thread, .313 Length, .176 Head Dia., .070 Head Height.



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TABLE III. PUSH-OUT AND TORQUE-OUT VALUES.

| T THREAD SIZE CLASS 2A | ALUMINUM ALLOY SHEET MATERIAL 3/ | | | COLD ROLLED STEEL SHEET MATERIAL 3/ | | |
|---|---|---|---|---|---|---|
| | SHEET THICKNESS (REF) 1/ | PUSH-OUT LOAD (LBS) MIN 2/ | TORQUE-OUT LOAD (IN LBS) MIN 2/ | SHEET THICKNESS (REF) 1/ | PUSH-OUT LOAD (LBS) MIN 2/ | TORQUE-OUT LOAD (IN LBS) MIN 2/ |
| .086-56 UNC | .020 | 40 | 4 | .023 | 85 | 4 |
| | .030 | 55 | 4 | .035 | 95 | 4 |
| .112-40 UNC | .020 | 80 | 8 | .023 | 160 | 8 |
| | .030 | 110 | 8 | .035 | 220 | 8 |
| .138-32 UNC | .020 | 90 | 12 | .023 | 160 | 13 |
| | .030 | 120 | 13 | .035 | 250 | 14 |
| .164-32 UNC | .020 | 110 | 12 | .023 | 170 | 20 |
| | .030 | 160 | 16 | .035 | 270 | 23 |
| .190-24 UNC | .020 | 120 | 13 | .023 | 190 | 21 |
| | .030 | 170 | 17 | .035 | 300 | 24 |
| .190-32 UNF | .020 | 120 | 13 | .023 | 190 | 21 |
| | .030 | 170 | 17 | .035 | 300 | 24 |

1/ Test panel thicknesses are optional between thread sizes, provided the corresponding load is maintained.

2/ The load values shown are the minimum non-destruct values that apply during testing.

3/ Sheet material hardness values shall not exceed 80HRB for use with carbon steel studs or exceed 70HRB for use with corrosion resistant steel studs.





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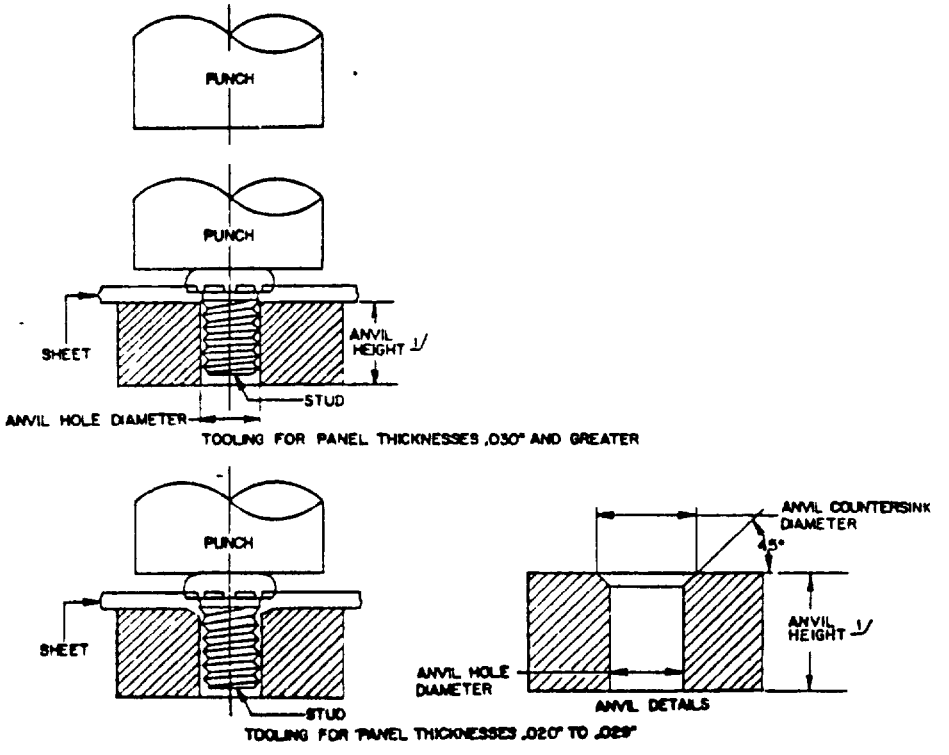
INSTALLATION PROCEDURES:

a. Any sheet thickness above the minimum indicated in Table IV for a specific thread size may be used.

b. Drill or punch hole in sheet material with diameter as shown in Table IV.

c. Locate shank of stud squarely in the hole in the mounting surface, with anvil below sheet (see Figure 2).

d. Apply force on punch sufficient only to embed ribs on head of stud into sheet, so that the underside of the stud head will be flush with the sheet surface.




1/ Anvil height minimum shall exceed "L" stud length such as to allow clearance for proper embedding of ribs on head of stud into sheet material.

FIGURE 2. GENERAL TOOLING SETUP.



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TABLE IV. INSTALLATION DIMENSIONS - (REFERENCE)

| T THREAD SIZE CLASS 2A | SHEET THICKNESS MIN | HOLE SIZE IN SHEET +.003 -.000 | CENTER OF HOLE TO SHEET EDGE MIN | NUT TIGHTENING TORQUE (IN LBS) MAX 1/ | ANVIL COUNTERSINK DIAMETER | ANVIL HOLE DIAMETER |
|---|---|---|---|---|---|---|
| .086-56 UNC | .020 | .085 | .187 | 2.3 | .110 to .114 | .087 to .090 |
| .112-40 UNC | .020 | .111 | .219 | 5.0 | .136 to .140 | .113 to .116 |
| .138-32 UNC | .020 | .137 | .250 | 9.0 | .162 to .166 | .139 to .142 |
| .164-32 UNC | .020 | .163 | .281 | 17.0 | .188 to .192 | .165 to .168 |
| .190-24 UNC | .020 | .189 | .281 | 24.0 | .216 to .220 | .191 to .194 |
| .190-32 UNF | .020 | .189 | .281 | 27.0 | .216 to .220 | .191 to .194 |

1/ This load is the limiting load when the nut is fastened to the stud at installation.





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*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2 DOCUMENT TITLE |
|---|---|
| MIL-S-63540/6 | STUD, SELF-LOCKING, NON-FLUSH, RIBBED CLINCH TYPE |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4 TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR<br>☐ USER<br>☐ MANUFACTURER<br>☐ OTHER *(Specify)* _____ |
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5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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